

Endogenous Event-Triggered Congestion Control for Urban Road Networks: A Sliding-Window Statistics Approach

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Abstract—Urban traffic congestion is stochastic and spatially propagating, while classical control approaches typically rely on plant models or external reference signals that are difficult to identify for large, heterogeneous urban networks. We propose an endogenous congestion controller that triggers path reallocation from sliding-window vehicle counts alone, with no external reference input, no offline training, and no dependence on network topology. The reallocation step is formalized as a constrained convex program in which the feedback gain K_p enters explicitly as a penalty coefficient; we establish existence, uniqueness, and non-expansiveness of this per-step allocation map, an explicit quantization-error bound for integer vehicle assignment, and a practical (bounded-perturbation) stability statement for the resulting closed loop, with non-divergence of the underlying microscopic traffic simulation verified empirically rather than asserted analytically. On a large-scale SUMO simulation of a cropped Philadelphia road network, enabling closed-loop actuation reduces both average travel time and cumulative congestion time simultaneously relative to a detection-only baseline, and achieves control performance comparable to or better than a GNN-enhanced baseline and two classical control baselines at a fraction of the per-step computation cost. An ablation over the feedback gain K_p further shows a graded, non-degenerate response below the network’s capacity-cap saturation point, and an exact convergence to identical outcomes above it, matching the saturation mechanism predicted by the underlying convex program’s KKT conditions.

Index Terms—Traffic congestion control; event-triggered control; constrained convex optimization; sliding window estimation; closed-loop feedback.

I. INTRODUCTION

Classical congestion management approaches—whether time-series, learning predictors or PID—share two fundamental limitations: prediction alone does not alleviate congestion, and model-based control relies on LTI assumptions rarely met in real traffic. These methods either chase forecasting accuracy without actuation, or track external references with identified plant models, remaining a gap between individual optimality (user equilibrium) and network-wide efficiency (system optimal) unaddressed.

This raises a central question: can we design a model-free, event-driven controller with rigorously scoped stability guarantees that mitigates network-wide congestion without sacrificing individual travel-time efficiency?

We answer this with an endogenous pre-allocation and feedback diversion system driven solely by sliding-window vehicle counts. At each 1-second step, arrival rates are estimated locally; diversion is actuated by solving a constrained convex program in which the feedback gain K_p enters as an

explicit penalty coefficient, whenever the estimated saturation exceeds a calibrated threshold. The theoretical foundation rests on two pillars: a bias-variance-bounded sliding-window estimator, and a convex-optimization allocator whose per-step map we prove exists, is unique, and is non-expansive in the risk estimate—together with an explicit account, via the program’s KKT conditions, of when and why the gain K_p saturates. These guarantees are scoped to the allocator’s own input-output behavior; the underlying physical traffic dynamics have no closed-form model and are verified empirically, not analytically, to remain bounded under the closed loop.

Structural distinctions from classical control:

- No reference tracking, no plant model, no derivative-based filtering;
- Event-triggered actuation: response delay determined by hardware, not system dynamics.

Contributions:

- 1) An endogenous diversion framework operating solely on sliding-window counts, with route reallocation posed as a constrained convex program in which the feedback gain K_p has a genuine, analytically traceable effect;
- 2) Existence, uniqueness, and non-expansiveness guarantees for the per-step allocation map, a deterministic quantization-error bound for integer vehicle assignment, and a practical (bounded-perturbation) stability statement for the resulting closed loop;
- 3) Simulation validation on a large-scale urban network showing that closed-loop actuation improves both travel time and congestion metrics simultaneously relative to a detection-only baseline, with no pre-training, no graph topology, and substantially lower per-step computation cost than a GNN-enhanced baseline;
- 4) An ablation over K_p that empirically confirms the theoretically predicted saturation behavior under tight capacity caps, explicitly accounting for why performance becomes insensitive to K_p once the capacity cap saturates.

II. RELATED WORK

A. Foundations of Poisson Traffic Modeling

Breiman established the Poisson tendency in traffic distribution, proving that under low-density conditions, vehicle counts on highway segments asymptotically follow a Poisson law—providing mathematical justification for treating vehicle arrivals as a Poisson process in urban networks [1]. Gordon and Newell embedded Poisson-distributed queues into a stochastic equilibrium framework, bridging queueing theory with traffic flow modeling [2]. Rényi formalized the Poisson

arrival model for divided highways under stationary conditions [3]. Lighthill and Whitham, and independently Richards, developed the kinematic wave theory [4]; Richards further revealed a critical threshold effect where perturbations amplify abruptly beyond a critical density—mirroring the congestion threshold θ employed in our feedback control formulation. These classical theories provide the statistical and physical foundation upon which our sliding-window estimation is built.

B. Conventional Forecasting and Their Limitations

Traditional traffic forecasting methods—including historical average, ARIMA/VAR, Kalman filtering, and their hybrids [5], [6]—are widely adopted but rely on stationarity or linearity assumptions that are frequently violated in urban traffic. Kernel-based approaches such as SVR introduce nonlinear mappings but incur prohibitive computational costs on large-scale networks and offer limited spatial modeling capabilities [7], [8]. Early neural network models, including LSTM and CNN-based architectures, improve short-term prediction accuracy but lack explicit spatial awareness or topological representation [9], [10].

More recently, graph neural networks (GNNs) have become the dominant paradigm for traffic forecasting by representing road networks as graphs, where nodes correspond to sensors or segments and edges capture spatial dependencies. DCRNN [7] formalized spatial dependency as a bidirectional diffusion process, while subsequent works have extended the framework along directions such as dynamic graph adaptation [11], [12], directional propagation modeling [13], and structural key-node identification [14], [15]. Transformer-based architectures have further pushed the frontier of long-sequence modeling [16], [17]. Despite their impressive predictive accuracy, these methods remain inherently forecasting-oriented: they provide estimates of future traffic states but offer no mechanism for translating those estimates into control actions. Prediction, however accurate, does not itself alleviate congestion—a gap that our framework directly addresses by embedding control logic into the statistical decision rule.

C. Event-Aware, Physics-Constrained, and Prediction-Control Integration

Recent studies have explored the integration of exogenous information and physical constraints into predictive models. FUSE-Traffic [18] uses large language models to extract event semantics and fuses them with GNN features via cross-attention. Shafiei et al. [19] integrate deep learning with traffic fundamental diagrams using a dual-stream architecture. Xu et al. [20] propose a dynamic risk field model using Gaussian kernels—conceptually parallel to our scalar congestion risk metric, though our formulation remains purely statistics-driven with no kernel parameterization. Sun et al. [21] proposed a stochastic congestion probability estimation framework but stopped short of incorporating feedback control.

The closest methodological relative to our work is the event-triggered control strategy for mixed traffic flow proposed by Wang et al. [22], which adjusts ACC vehicle time gaps to suppress stop-and-go waves based on the ARZ PDE model. Their

small-gain triggering ensures exponential stability without Zeno behavior. Nevertheless, key distinctions remain: Wang et al. rely on continuous PDE dynamics requiring explicit model knowledge, whereas our approach operates on discrete event counts with no plant model; their analysis is tailored to ACC-equipped homogeneous freeways, while ours applies to arbitrary urban networks with heterogeneous compositions; and their triggering condition depends on deviations from a reference equilibrium, whereas ours defines the threshold directly on empirical saturation statistics, eliminating reference tracking altogether.

D. Research Gaps and Positioning

Despite substantial progress in forecasting accuracy, the literature remains constrained by two interconnected gaps. First, the vast majority of data-driven methods are prediction-oriented—control, if considered, is treated as a post-processing step rather than an integral part of the decision loop. Second, even when actuation is present, classical controllers rely on LTI models and external references, while emerging event-triggered strategies are tailored to specific PDE-based scenarios (e.g., ACC-equipped freeways), leaving general urban networks with stochastic arrivals unaddressed. Critically, neither line of work quantifies the trade-off between user equilibrium (individual optimality) and system optimality (network-wide efficiency)—a gap that fundamentally separates forecasting accuracy from control efficacy.

This paper directly addresses these gaps through an endogenous framework that: (i) embeds control logic into the statistical decision rule, closing the prediction-control loop at the algorithmic level; (ii) operates without plant models, external references, or derivative-based filtering; (iii) applies to general urban topologies with rigorous contraction-based stability guarantees; and (iv) explicitly characterizes the UE-SO trade-off through ablation analysis. In doing so, we shift the emphasis from *forecasting what will happen* to *deciding what to do about it*—a distinction we argue is fundamental to real-time traffic management.

III. THEORETICAL FRAMEWORK OF THE ENDOGENOUS SYSTEM

This section establishes the theoretical foundations of the proposed endogenous congestion control framework. To avoid conflating distinct notions of convergence, we state explicitly, at the outset, the four analytical layers addressed below and the precise claim made (or deliberately not made) at each layer:

- **L1 (Physical traffic layer):** vehicle queues, densities, and travel times, generated by SUMO’s microscopic car-following and lane-changing simulation. We do *not* derive an analytic global-stability guarantee for this layer; non-divergence and recovery behavior are instead verified empirically via simulation (Section V).
- **L2 (Risk-estimation layer):** the sliding-window/EWMA arrival-rate estimator $\hat{\lambda}_e$. We establish a bias-variance bound for this layer (Theorem 1) and a unit-gain property of its normalized form (Lemma 1).

- **L3 (Continuous path-flow control layer):** the projection-based optimization that produces the continuous allocation u . We establish existence, uniqueness, and non-expansiveness (Lipschitz continuity in the risk estimate) of this per-step map (Theorem 2), together with an explicit account of why the feedback gain K_p appears to saturate empirically (Proposition 1). This is strictly a statement about the allocator's input-output map, not about the closed-loop physical traffic system.
- **L4 (Integer execution layer):** the discretized vehicle-path assignment z obtained from u by deterministic rounding. We derive an exact quantization error bound $\|z - u\|_\infty$ (Proposition 2).

The remainder of this section proceeds as follows. Section III-A develops the statistical foundations of the arrival-rate estimator (L2). Section III-B develops the continuous allocation layer as a constrained convex optimization in which K_p enters explicitly as a penalty coefficient, establishes existence/uniqueness/non-expansiveness of the resulting map (L3), and explains the empirically observed insensitivity of Table II to K_p via KKT complementary slackness rather than treating it as an unexplained coincidence. Section III-C bounds the quantization error introduced by integer vehicle assignment (L4). Section III-D synthesizes L1–L4 into a practical-stability statement for the closed loop, clarifying precisely why boundedness and non-expansiveness alone do not, by themselves, imply global asymptotic stability of the physical system.

A. Statistical Foundations: Poisson Process and Sliding Window

1) *Poisson Arrivals as a Baseline Estimator, Not a Physical Law:* We adopt a Poisson process as a *baseline (nominal) model* for vehicle arrivals to a road segment. This model is used exclusively to motivate the design and calibrate the bias-variance behavior of the sliding-window estimator introduced below (Theorem 1); it is not asserted as an exact physical description of urban traffic arrivals. Under the classical Poisson limit theorem, if the number of vehicles N in the network is large while the probability p that an individual vehicle selects a given segment within a short interval is small, with $Np = \lambda$ finite, the arrival count process is well approximated by

$$P(N(t + \Delta t) - N(t) = k) = \frac{(\lambda(t) \cdot \Delta t)^k}{k!} e^{-\lambda(t)\Delta t}. \quad (1)$$

We next examine the three defining axioms of this baseline model and state their physical scope of validity precisely, rather than treating them as exact:

- **Ordinariness:** the exact Poisson requirement is $P(N(\Delta t) \geq 2) = o(\Delta t)$ as $\Delta t \rightarrow 0$. In signalized urban networks this is *not* exactly true: traffic signals discharge vehicles in platoons at cycle boundaries, producing locally bursty, over-dispersed arrivals that violate ordinarity at sub-second resolution. We do not claim that two vehicles cannot arrive within an infinitesimal interval; rather, at our 1-second sampling resolution, the platooning effect is treated as a source of over-dispersion

whose magnitude is quantified empirically (Section V) via a dispersion-index (Fano-factor) diagnostic comparing the empirical variance-to-mean ratio of second-level arrival counts against the Poisson baseline of unity.

- **Independent increments:** arrival counts over disjoint intervals are assumed independent under the baseline model. Physically, signal-cycle discharge induces short-range temporal correlation, and route diversion induces spatial correlation across the network; neither is exactly zero. These correlations are exactly the phenomena the dispersion diagnostic above is designed to detect and report, rather than assume away.
- **Stationary increments:** the baseline model assumes a time-invariant rate. Tidal traffic patterns make $\lambda(t)$ time-varying at the scale of tens of minutes, so global stationarity does not hold; local stationarity is only approximately valid within a short window, which motivates the sliding-window construction below.

Scope of the Poisson Assumption. The Poisson model is used only to inform the bias-variance analysis of the estimator in Theorem 1; none of the results in Sections III-B through III-D (the continuous allocation map, its non-expansiveness, or the quantization bound) depend on arrivals being exactly Poisson. Those results depend only on the algebraic structure of the projection-based optimization and the deterministic rounding rule, and therefore hold regardless of the true arrival distribution. Where the Poisson assumption is load-bearing (Theorem 1), we restrict the claim to first- and second-moment (bias/variance) behavior, which is far more robust to distributional misspecification than the full distributional form of Eq. (1).

2) *Sliding-Window Estimation: Bias-Variance Trade-off, Not Optimality:* The sliding window of length T_w provides an online estimate of the local arrival rate. We state its statistical properties precisely, without over-claiming optimality of any specific window length.

Theorem 1 (Bias-Variance Trade-off of the Sliding-Window Estimator): Within the sliding window $[k - T_w, k]$, the estimator $\hat{\lambda}_e(k) = N_e(k)/T_w$, where $N_e(k)$ is the measured arrival count accumulated over the window, satisfies

$$\begin{aligned} \mathbb{E}[\hat{\lambda}_e(k)] &= \lambda_e(k) + O(T_w \cdot \sup |\lambda'_e|), \\ \text{Var}(\hat{\lambda}_e(k)) &= \frac{\lambda_e(k)}{T_w}. \end{aligned} \quad (2)$$

Proof sketch. The bias term follows from a first-order Taylor expansion of $\lambda_e(\cdot)$ within the window under local homogeneity; the variance follows from the Poisson (or, more generally, any process with variance proportional to its mean) baseline. ■

Equation (2) exhibits the classical bias-variance trade-off of window-based rate estimation: increasing T_w reduces variance at rate $O(1/T_w)$ but increases bias whenever $\lambda_e(\cdot)$ is non-constant over the window. **We do not claim that $T_w = 60$ s is variance-optimal, bias-optimal, or MSE-optimal in any formal sense.** It is an engineering choice motivated only by the observation that the characteristic time scale of tidal

urban traffic (tens of minutes) is much larger than 60 s, so that the local-homogeneity approximation underlying Eq. (2) is reasonable at this scale. Section V reports a sensitivity sweep over $T_w \in \{15, 30, 60, 120\}$ seconds on a held-out validation split, computing the empirical mean squared error $\text{MSE}(T_w) = \text{Bias}^2(T_w) + \text{Var}(T_w)$ directly from simulated arrival traces, so that the choice of T_w is substantiated empirically rather than asserted as theoretically optimal.

3) *Normalized Estimator and the Collapse of the Naive Gain*: The algorithmic implementation (Section IV-B) realizes the sliding-window estimate via an exponentially decayed accumulation with decay factor α . A dimensionally consistent, normalized estimator is

$$\hat{\lambda}_e(k) = \frac{\sum_{\tau=0}^{T_w-1} \alpha^\tau n_e(k-\tau)}{\Delta t \cdot \sum_{\tau=0}^{T_w-1} \alpha^\tau}, \quad (3)$$

where Δt is the sampling interval (1 s) and $n_e(k-\tau)$ is the *measured* arrival count at lag τ (the measured/predicted distinction is made precise in Section IV-B1). Normalizing by $\Delta t \sum_{\tau=0}^{T_w-1} \alpha^\tau$ makes $\hat{\lambda}_e(k)$ carry units of veh/s, directly comparable to the normalized capacity $c_e = C_e/3600$ (veh/s) used in the risk computation of Section IV-B3. The decay factor $\alpha = 0.95$ is retained as a data-informed, pragmatic choice: the effective memory length $1/(1-\alpha) = 20$ steps is of the same order as the empirically observed autocorrelation time scale (15–25 s) of second-level arrival rates in urban networks; we do not claim this is the MMSE gain for a specific AR(1) model, only that it is a reasonable match to the observed correlation structure.

Lemma 1 (Normalization Collapses the Estimator Gain to Unity): Let $\hat{A}$ denote the linear map in Eq. (3) from the bounded arrival sequence $\{n_e(k-\tau)\}_{\tau=0}^{T_w-1}$ to $\hat{\lambda}_e(k)$. Because the weights α^τ are normalized by their own sum $\sum_{\tau=0}^{T_w-1} \alpha^\tau$, the induced ℓ_∞ -gain of $\hat{A}$ with respect to a bounded perturbation of the arrival sequence is

$$\|\hat{A}\|_\infty = \frac{\sum_{\tau=0}^{T_w-1} \alpha^\tau}{\Delta t \sum_{\tau=0}^{T_w-1} \alpha^\tau} \cdot \Delta t = 1.$$

This is a substantive consequence of the normalization, not a cosmetic one: an *un-normalized* accumulation (weights α^τ without division by $\sum_{\tau=0}^{T_w-1} \alpha^\tau$) would instead yield a gain factor $\sum_{\tau=0}^{T_w-1} \alpha^\tau = \frac{1-\alpha^{T_w}}{1-\alpha} \approx 19.08$ for $\alpha = 0.95$, $T_w = 60$. Any downstream argument that multiplies this un-normalized gain by a feedback coefficient would silently inflate the effective loop gain by a factor of ≈ 19.08 relative to the normalized estimator, since the un-normalized gain is not invariant to the choice of normalization. We therefore do not carry the estimator's gain forward into the contraction argument of Section III-B at all: with the normalization of Eq. (3), $\|\hat{A}\|_\infty = 1$, and the sole source of feedback gain in the system is the coefficient K_p introduced explicitly in the optimization problem below, not an implicit multiplicative interaction with the estimator's window length.

B. Continuous Path-Flow Allocation: Existence, Uniqueness, and Non-Expansiveness (Path B)

We define the allocator itself as the solution of an explicit, strongly convex optimization problem in which K_p is a genuine penalty coefficient, so that its value has a direct, analytically traceable effect on the resulting allocation.

1) *The Allocator as a Constrained Convex Program*: At each control step k and for each origin-destination pair with candidate paths $p = 1, \dots, m$ and demand D , let $u^0(k) \in \mathbb{R}^m$ be a feasible reference allocation (e.g., the free-flow-weighted baseline of Section IV-C1) and let $\rho_p(k)$ denote the congestion risk score of path p at time k (Section IV-C2 defines $\rho_p(k)$ explicitly in terms of the edge-level risk $r_e(k)$). The continuous allocation $u(k)$ is defined as the unique solution of

$$u(k) = \arg \min_{u \in \mathcal{U}(D)} \frac{1}{2} \|u - u^0(k)\|_2^2 + K_p \sum_{p=1}^m \rho_p(k) u_p, \quad (4)$$

where the feasible set $\mathcal{U}(D) = \{u \in \mathbb{R}^m : \sum_p u_p = D, 0 \leq u_p \leq L_p \forall p\}$ encodes demand conservation and path capacity limits L_p . Here K_p is a genuine coefficient in the objective: it directly penalizes allocating flow to high-risk paths, and its value has a direct, analytically traceable effect on the optimizer.

Theorem 2 (Existence, Uniqueness, and Non-Expansiveness of the Allocator Map): Assume $\mathcal{U}(D)$ is nonempty (i.e., $D \leq \sum_p L_p$; the infeasible case is handled by the elastic relaxation of Section IV-C3). Then:

- (i) Problem (4) has a unique minimizer $u^*(k)$, since the objective is strongly convex (its Hessian is the identity) and $\mathcal{U}(D)$ is a nonempty, closed, convex, bounded set.
- (ii) By the KKT stationarity condition, $u^*(k) = \Pi_{\mathcal{U}(D)}(u^0(k) - K_p \rho(k))$, where $\Pi_{\mathcal{U}(D)}$ is the Euclidean projection onto $\mathcal{U}(D)$.
- (iii) Because Euclidean projection onto a convex set is non-expansive, $\|\Pi_{\mathcal{U}(D)}(x) - \Pi_{\mathcal{U}(D)}(y)\|_2 \leq \|x - y\|_2$, the map $r \mapsto u^*(r)$ is Lipschitz with constant at most $K_p \cdot L_\rho$, where L_ρ is a Lipschitz constant of the risk-to-cost map $r \mapsto \rho(r)$ used in Section IV-C2.

Scope of Theorem 2. This result establishes existence, uniqueness, and Lipschitz continuity of the *per-step allocator's own input-output map*—Path B, in the terminology of the introduction to this section. It is not a claim about the physical traffic system's stability. For completeness, we record the general form of the closed-loop system that would need to be analyzed to obtain a physical-layer (L1) contraction guarantee:

$$\begin{aligned} x_{k+1} &= \Phi(x_k, u_k, d_k), \\ r_k &= H(x_k), \\ u_k &= \Pi_{\mathcal{U}(D)}[u_k^0 - K_p \Psi(r_k, \theta)]. \end{aligned} \quad (5)$$

where Φ is the physical traffic dynamics, d_k is exogenous demand, H maps state to risk, and Ψ is the risk-to-penalty non-linearity. A sufficient contraction condition for the *full* closed loop would take the form $L_{\text{closed}} \leq L_x + L_u \cdot K_p \cdot L_\Psi \cdot L_H < 1$, where L_x , L_u , L_Ψ , L_H are Lipschitz constants of Φ with respect to x_k and u_k , of Ψ , and of H , respectively. Because Φ is realized by SUMO's microscopic car-following and lane-changing model and has no closed-form expression, L_x cannot

be derived analytically. **We therefore explicitly do not claim** $L_{\text{closed}} < 1$ **or any global contraction of the physical system**, and restrict the analytic guarantee to Theorem 2 (Path B); non-divergence of the physical layer is instead verified empirically in Section V.

2) *Saturation-Induced K_p -Insensitivity via Complementary Slackness*: **Proposition 1 (KKT Saturation Explanation)**: Consider the KKT conditions of problem (4). For a path p whose box constraint is active at the optimum ($u_p^* \in \{0, L_p\}$), complementary slackness gives a nonzero multiplier absorbing any further increase in the penalty term $K_p \rho_p(k)$: once $K_p \rho_p(k)$ is large enough to push $u_p^0 - K_p \rho_p(k)$ outside $[0, L_p]$, the projection in Theorem 2(ii) clips u_p^* to the corresponding bound *regardless of how much further K_p increases*, because the projection of any value below 0 (resp. above L_p) is 0 (resp. L_p) irrespective of magnitude. Consequently, once K_p is large enough that the congestion-penalized paths in a given OD group are already saturated at their lower bounds and the non-congested paths absorb the redistributed demand up to their own capacity or the $\gamma_{\max}$ cap (Section IV-C2), further increases in K_p produce *no additional change* in $u^*(k)$. This is the precise mechanism—rooted in the KKT complementary slackness condition of a well-posed convex program, not an unexplained empirical coincidence—underlying the near-identical outcomes observed for $K_p \in [0.02, 1.0]$ in Table II: the saturation boundary imposed by $\mathcal{U}(D)$ and $\gamma_{\max}$ dominates the allocator’s behavior once K_p exceeds the (typically small) threshold needed to saturate the congested paths.

C. Quantization Layer: Error Bound for Integer Vehicle Assignment

The continuous allocation $u(k)$ must be realized as an integer number of vehicles per path. We bound the resulting discretization error explicitly, rather than leaving it unquantified.

Proposition 2 (Quantization Error Bound): Let $u \in \mathbb{R}_{\geq 0}^m$ satisfy $\sum_i u_i = D$ with D a nonnegative integer. The vehicle-path assignment rule used in Section IV-D (cumulative sums $c_i = \sum_{k \leq i} u_k$, assigning vehicle $j \in \{0, \dots, D-1\}$ to the first path with $c_i > j + 0.5$) implements the *largest-remainder (Hamilton/Webster) apportionment method*: each path’s integer quota z_i is $\lfloor u_i \rfloor$ plus one unit awarded to the paths with the largest fractional remainders, until the total D is exactly matched. By the classical quota property of the largest-remainder method,

$$\|z - u\|_{\infty} < 1, \quad (6)$$

independent of D , m , or the specific values of u . This deterministic bound is used in Section III-D as the magnitude of the bounded perturbation contributed by integer execution (L4) to the closed-loop practical-stability statement.

D. Practical Stability of the Closed Loop

We now synthesize L1–L4 into a single, carefully scoped stability statement. Two natural but insufficient claims are worth ruling out explicitly at this point: boundedness alone does not imply asymptotic stability, and non-expansiveness

alone does not preclude the possibility of multiple fixed points or periodic orbits in general. We therefore state only the following practical-stability result, restricted to what is actually established by Theorems 2 and 3 below.

Define the Lyapunov candidate $V(u_k) = \|u_k - u^*\|_{\infty}$ relative to a reference fixed point u^* of the per-step allocator map at a frozen risk vector r^* (i.e., $u^* = \Pi_{\mathcal{U}(D)}(u^0 - K_p \rho(r^*))$ held fixed). By Theorem 2(iii), if the risk vector were exactly constant at r^* across steps, V would be exactly constant (the allocator has no internal recursion in u at fixed r); the actual discrete-time dynamics of interest are those of r_k itself, driven by the physical layer Φ in Eq. (5), which we do not have in closed form. We therefore do not claim an exact contraction $V(u_{k+1}) \leq L V(u_k)$ for a specific numerical L ; as shown in Lemma 1, a numerical loop-gain figure of this kind is only meaningful once the estimator is properly normalized, and no single scalar contraction rate for the full closed loop is asserted here.

Theorem 3 (Practical Stability Under Bounded Perturbations): Let $w_k = (w_k^{\text{quant}}, w_k^{\text{demand}})$ collect the quantization error bounded by Proposition 2 ($\|w_k^{\text{quant}}\|_{\infty} < 1$) and stochastic demand fluctuations around the nominal arrival rate. Suppose (verified empirically in Section V, not claimed analytically here) that the physical-layer trajectory $r_k = H(x_k)$ satisfies, for some class- $\mathcal{KL}$ function β and class- $\mathcal{K}$ function γ ,

$$\|r_k - r^{\text{ref}}\|_{\infty} \leq \beta(\|r_0 - r^{\text{ref}}\|_{\infty}, k) + \gamma\left(\sup_{j \leq k} \|w_j\|_{\infty}\right),$$

i.e., the closed loop is input-to-state practically stable with respect to a bounded reference neighborhood. Then the deviation of r_k from r^{ref} is bounded by a term that decays with k plus a term proportional to the bounded quantization and demand perturbations, rather than converging exactly to a single point. This is the sense in which we describe the closed loop as *practically stable*: the guarantee is a bounded neighborhood of convergence under bounded disturbances, not exact asymptotic convergence to a unique fixed point of the full physical system. Section V verifies this bound empirically by tracking r_k across simulation runs under bounded quantization and demand-fluctuation perturbations and confirming non-divergence and recovery, since β and γ are not analytically available for the SUMO-simulated physical layer.

Physical interpretation: $V(r) = \|r - r^{\text{ref}}\|_{\infty}$ measures the maximum saturation deviation from a reference operating point—the “peak congestion” of the network relative to that reference. Theorem 3 states that this peak congestion remains within a bounded neighborhood that shrinks with time and grows only with the magnitude of quantization and demand disturbances; it does not assert monotonic decrease at every step, nor an exponential rate with a specific numerical constant, since those would require the physical-layer Lipschitz constant L_x in Eq. (5), which is not available in closed form for a microscopic traffic simulator. This is a deliberately conservative guarantee, weaker than an exact global exponential stability claim, but one that holds without requiring an analytic model of the physical layer.

IV. METHODOLOGY

This section presents the complete algorithmic implementation of the endogenous pre-allocation and feedback diversion system, together with an explicit statement of the physical traffic-loading model that generates ground-truth vehicle movement (Section IV-A). The architecture consists of three tightly coupled components: a physical loading model that governs how vehicles actually move (Section IV-A), a sliding-window congestion tracker that maintains statistical estimates of link saturation from measured and predicted arrivals (Section IV-B), and a route allocator that maps vehicles to paths by solving the constrained convex program of Section III-B (Section IV-C–IV-D). All components operate at a 1-second discrete time step.

A. Traffic Loading Model and Simulation Protocol

1) *Physical Layer: Microscopic Simulation as Ground Truth:* Vehicle movement, travel time, queueing, and capacity constraints are generated entirely by `TRAFFIC_LOADING_MODEL = SUMO_MICROSCOPIC_CARFOLLOWING`: SUMO's car-following and lane-changing microsimulation constitutes the physical (L1) layer of this work. We do not additionally employ an analytic macroscopic loading model (e.g., the cell transmission model) to generate ground-truth flows; the endogenous tracker and allocator described in Sections IV-B–IV-D act only as a routing controller layered on top of SUMO via `traci.vehicle.setRoute` calls, and do not themselves move vehicles.

For interpretive and monitoring purposes, we additionally record an aggregate point-queue-style bookkeeping quantity for each edge e ,

$$\begin{aligned} n_e(k+1) &= n_e(k) + a_e(k) - d_e(k), \\ d_e(k) &= \min(n_e(k) + a_e(k), \mu_e \cdot \Delta t). \end{aligned} \quad (7)$$

where $a_e(k)$ is the number of vehicles physically entering edge e at step k , $d_e(k)$ is the number discharged (bounded by the saturation flow rate μ_e , veh/s), and $n_e(k)$ is the resulting queue occupancy. Equation (7) is a monitoring proxy used to report queue-length diagnostics (Section V); it is not the mechanism by which vehicle trajectories, speeds, or travel times are computed—those are produced directly by SUMO's microsimulation, which additionally captures spillback, gap-acceptance, and car-following effects that Eq. (7) does not model.

2) *Simulation Protocol: Warm-up, Loading, and Clearance:* Each experiment run follows a three-phase protocol:

- **Warm-up** (`WARMUP_DURATION`): the network is populated with initial traffic and the controller operates normally, but metrics are not yet recorded, so that reported statistics are not biased by an empty-network transient.
- **Loading** (`sim_duration`): the demand profile is injected and metrics are recorded.
- **Clearance** (`CLEARANCE_DURATION`): demand injection stops but the simulation continues until either no vehicles remain on the network or the clearance duration

elapses, so that in-progress trips are allowed to complete rather than being truncated at the loading horizon.

At the end of the clearance phase we report $\text{incomplete_ratio} = N_{\text{incomplete}}/N_{\text{departed}}$, the fraction of departed vehicles that never produce a completed trip record (`tripinfo`) by the end of the clearance window. This quantity is reported alongside every experiment so that average-trip-time statistics can be interpreted with explicit knowledge of how much of the demand they actually reflect.

B. Sliding-Window Congestion Tracker: The Core of Endogenous Sensing

1) *Measured vs. Predicted Arrivals: A Strict Notational Separation:* Because the tracker is endogenous—it has no external sensor independent of its own routing decisions—we strictly separate two classes of quantities:

- $n_e(k)$: the *measured* arrival count on edge e at step k , obtained after the fact from realized simulation state (e.g., via `traci.edge.getLastStepVehicleNumber` or an equivalent measured count).
- $\tilde{n}_e(k + \tau \mid k)$: the *predicted* future arrival at lag τ , computed at step k from vehicles just assigned to paths whose free-flow travel-time offset places them on edge e exactly τ steps ahead.

The circular buffer $\mathbf{W} \in \mathbb{R}^{N_e \times T_w}$ populated by `update_allocation` is necessarily filled with predicted quantities $\tilde{n}_e(k + \tau \mid k)$, not measured quantities $n_e(k)$, since the tracker only knows what it has itself just assigned. This self-referential prediction accumulates error whenever realized traversal times deviate from the free-flow assumption (e.g., under congestion-induced slowdown), and is stated explicitly here rather than described as “real-time observation.”

Online correction. To bound this self-referential error, once a vehicle's realized arrival at edge e becomes measurable at step k , the buffer's zero-lag slot is overwritten with the measured value, $W_{e,0} \leftarrow n_e(k)$, replacing whatever value was previously predicted for this step at an earlier time $k - \tau$. Only the strictly future entries ($\tau \geq 1$) of the buffer remain prediction-based at any given time, which bounds the self-referential prediction horizon to at most $T_w - 1$ steps rather than allowing compounding error to accumulate indefinitely.

2) *Circular Buffer Data Structure:* Let N_e denote the number of road segments and $T_w = 60$ the sliding-window length in time steps, matching the bias-variance trade-off of Theorem 1. The buffer $\mathbf{W}$ is maintained with $O(1)$ insertion via a write pointer `current_time_idx` and a `step_forward()` operation that advances the pointer and clears the incoming slot, discarding data older than T_w seconds. This provides a finite, bounded memory footprint of $N_e \times T_w$ floating-point values, independent of simulation duration, consistent with the local-homogeneity requirement of Theorem 1.

3) *Write-Time Recording and Read-Time Decay:* The decay factor α^τ is applied *exactly once, at read time*, matching Eq. (3): the buffer stores undecayed contributions, and decay is applied only when the accumulated sum is read, avoiding a compounded $\alpha^{2\tau}$ effect. The capacity term C_e (veh/h) is likewise converted to veh/s ($c_e = C_e/3600$) before being

compared against the estimator, so that the risk computation below is dimensionally consistent throughout. At write time, when a vehicle is assigned to edge e with expected offset $\tau \in [0, T_w - 1]$ and weight w , the (undecayed) contribution is recorded as

$$W_{e,\tau} \leftarrow W_{e,\tau} + w,$$

and the decay factor α^τ is applied only once, at read time, in the risk computation below.

4) *Threshold-Triggered Congestion Detection*: At each 1-second step, the tracker computes the normalized arrival-rate estimate and the resulting congestion risk as

$$\begin{aligned} \hat{\lambda}_e(k) &= \frac{\sum_{\tau=0}^{T_w-1} \alpha^\tau W_{e,\tau}}{\Delta t \sum_{\tau=0}^{T_w-1} \alpha^\tau}, \\ r_e(k) &= \frac{\hat{\lambda}_e(k)}{c_e}, \\ c_e &= \frac{C_e}{3600} \text{ (veh/s)}, \end{aligned} \quad (8)$$

directly matching the normalized estimator of Eq. (3) and maintaining consistent units throughout. This dimensionless saturation metric is comparable across edges of different physical capacities.

The trigger condition is $\text{congested}(e) = \mathbf{1}[r_e(k) \geq \theta_e]$. The threshold θ_e is the only externally set parameter in the framework and is calibrated, not hand-tuned: we designate disjoint sets of simulation seeds CALIBRATION_SEEDS, VALIDATION_SEEDS, and TEST_SEEDS, and set

$$\theta_e = \text{Quantile}_q(\{r_e(k) : k \text{ over CALIBRATION_SEEDS}\}),$$

where the quantile level q is selected by evaluating detection performance (Section IV-C2's precision/recall metrics) on VALIDATION_SEEDS only; TEST_SEEDS are held out and evaluated exactly once, with θ_e frozen. This formulation employs a genuine empirical quantile computed from calibration data, in contrast to a min-max rescaling rule of the form $\theta_e = r_{\min} + q \cdot (r_{\max} - r_{\min})$, and prevents threshold tuning on the test split.

The tracker's operations have complexity $O(K)$ for recording allocations (K : vehicles) and $O(N_e \cdot T_w)$ for risk computation, with $O(N_e \cdot T_w)$ memory.

C. Path Allocation with Congestion Feedback

The route allocator solves the convex program of Eq. (4) at each control step, translating the tracker's congestion estimates into control actions. We first define terminology precisely to align with standard traffic assignment nomenclature and avoid conceptual ambiguity.

1) *Path Precomputation and Terminology*: For each origin-destination pair (o, d) , candidate paths $\mathcal{P}_{od} = \{p_1, \dots, p_m\}$ are precomputed with edge sequences $\text{edges}(p)$, traversal offsets $\text{offsets}(p)$, and priority weight $w_p = 1/\text{free_flow_time}(p)$. The reference allocation:

$$u_i^0 = \min \left(D \cdot \frac{w_i}{\sum_j w_j}, L_i \right) \quad (9)$$

with remaining demand redistributed proportionally among unsaturated paths is used as the baseline point $u^0(k)$ in Eq. (4). We refer to the $K_p = 0$ condition (where the penalty term in Eq. (4) vanishes and $u(k) = u^0(k)$) as the **free-flow-weighted baseline allocation**, and to the $K_p > 0$ condition as the **resilience-oriented allocation**. We explicitly do *not* call these "Wardrop User Equilibrium" or "System Optimum": UE requires equalized realized generalized cost across used paths per OD pair, and SO requires minimization of total system travel time, neither of which is verified or claimed for either condition here. A rigorous UE/SO comparison would require an independent standard DTA/UE/SO solver reporting a Wardrop gap, which is outside the scope of this paper's controller-focused contribution.

2) *Path Risk, the Convex Program, and the Scope of $\gamma_{\max}$* : The path-level risk used in Eq. (4) is defined as the maximum edge-level risk along the path, $\rho_p(k) = \max_{e \in \text{edges}(p)} r_e(k)$, so that a path is only penalized to the extent that some edge on it is at risk of congestion. The allocator applies the following per-OD-pair-scoped congestion constraint: letting D_{od} be the demand of a given OD pair and $D_{od,\text{cong}}(k) = \sum_{p: \rho_p(k) \geq \theta} u_p(k)$ the flow it places on paths flagged congested, we require $D_{od,\text{cong}}(k) \leq D_{od} \cdot \gamma_{\max}$ with $\gamma_{\max} = 0.4$ as an additional linear constraint folded into $\mathcal{U}(D)$ for that OD pair.

We state explicitly that $\gamma_{\max}$ is applied per OD pair, not as a single network-wide aggregate cap. Algorithm 1 (step 4 below) enforces this constraint within each OD pair's own convex program, and the resulting guarantee—at least 60% of each OD pair's own demand avoids paths flagged as congested—is likewise a per-OD-pair, not network-wide, guarantee.

3) *Iterative Water-Filling Solution*: Problem (4), restricted to a single OD pair's box-and-simplex feasible set, is solved by iterative water-filling: initialize the active (unsaturated) path set $\mathcal{A} = \{1, \dots, m\}$; at each iteration, solve for the water level μ such that $\sum_{i \in \mathcal{A}} \text{clip}(u_i^0 - K_p \rho_i + \mu, 0, L_i) = D - \sum_{i \notin \mathcal{A}} u_i$ by bisection on μ ; any path whose clipped value hits 0 or L_i is removed from $\mathcal{A}$ and fixed at that bound; repeat until no further path saturates in an iteration ($R = 0$) or $\mathcal{A} = \emptyset$. This iterative procedure refines the single-shot redistribution scheme to convergence, eliminating residual unallocated demand that can arise when multiple paths saturate simultaneously, and guarantees $\sum_i u_i = D$ exactly whenever $\mathcal{U}(D)$ is nonempty. **Infeasibility fallback.** When $\mathcal{U}(D)$ is empty—either because $D > \sum_i L_i$ or because the $\gamma_{\max}$ constraint cannot be met without violating a capacity bound—we solve an elastic relaxation

$$\begin{aligned} \min_{u, s \geq 0} \quad & \frac{1}{2} \|u - u^0\|_2^2 + K_p \sum_p \rho_p u_p + M \sum_i s_i, \\ \text{s.t.} \quad & \sum_i u_i = D, \quad 0 \leq u_i \leq L_i + s_i, \end{aligned} \quad (10)$$

with a large penalty M on the slack variables s_i , producing a minimum-constraint-violation solution. The unmet diversion demand, $\sum_i s_i$ at the optimum, is recorded and reported explicitly (unmet_demand metric) rather than silently absorbed.

4) *Vehicle-Path Assignment (Quantization)*: The continuous allocation $u(k)$ is discretized into an integer vehicle count per path via cumulative sums $c_i = \sum_{k \leq i} u_k$, assigning vehicle j to the first path with $c_i > j + 0.5$. As established in Proposition 2, this is the largest-remainder apportionment method and satisfies the deterministic quota bound $\|z - u\|_\infty < 1$; this bound is what is carried into the practical-stability statement of Theorem 3 as the quantization perturbation w_k^{quant} .

D. Summary of the Endogenous Feedback Loop

Algorithm 1 summarizes the complete endogenous feedback loop in seven explicit steps: reading true measured detection values, causal window updates that tolerate missing data, explicit risk/capacity construction, constrained optimization with an infeasibility fallback, quantization with conservation correction, a cross-OD-pair edge capacity check, and advancement of the physical loading model.

Algorithm 1 Endogenous Feedback Loop (One Time Step)

- 1: **Step 1 (True detection)**: Read measured arrival counts $n_e(k)$ from the physical layer (Section IV-A), not only self-recorded predictions $\tilde{n}_e(k | k - \tau)$; overwrite $W_{e,0}$ with $n_e(k)$ (online correction, Section IV-B1).
 - 2: **Step 2 (Causal window update)**: *tracker.step_forward()*; if a measurement is missing for a given edge/step (sensor dropout), hold the last valid value rather than treating the gap as zero arrivals.
 - 3: **Step 3 (Risk and capacity construction)**: Compute $\hat{\lambda}_e(k)$, $r_e(k)$ via (8); compute path risks $\rho_p(k) = \max_{e \in p} r_e(k)$ and remaining path capacities L_p .
 - 4: **for** each OD pair with demand $D_{od} > 0$ **do**
 - 5: **Step 4 (Constrained optimization)**: Solve (4) via iterative water-filling (Section IV-C3) subject to the per-OD $\gamma_{\max}$ constraint; if infeasible, solve the elastic relaxation and record unmet demand.
 - 6: **Step 5 (Quantization and conservation correction)**: Round $u(k)$ to integer vehicle counts $z(k)$ via largest-remainder assignment (Proposition 2); the rounding rule guarantees $\sum_i z_i = D_{od}$ exactly.
 - 7: **end for**
 - 8: **Step 6 (Cross-OD capacity check)**: Aggregate assigned flows across all OD pairs onto each edge; if any edge's aggregate assignment exceeds its physical capacity, reconcile via the elastic relaxation of the affected OD pairs.
 - 9: **Step 7 (Advance physical layer)**: Advance the SUMO loading model one step (Section IV-A); record the resulting measured state $n_e(k + 1)$, realized travel times, and queue/halting diagnostics for the next iteration.
 - 10: Return assignments.
-

This loop retains the endogenous character of the design—the system feeds on its own measured and predicted state to anticipate congestion, actuating only when the threshold is crossed—and ensures full consistency between the theoretical formulation and the algorithmic implementation: K_p explicitly parameterizes the optimization solved at Step 4, and Steps 5–7 make explicit the integer-execution and

physical-loading components that the theoretical framework's L4 and L1 layers, respectively, refer to.

E. Connection to Theoretical Guarantees

The algorithm implements the theoretical framework of Section III as follows:

- The normalized risk computation in Eq. (8) directly realizes the normalized estimator of Eq. (3), with consistent normalization applied in both the theoretical analysis and the algorithmic implementation (Sections III-A3 and IV-B3–IV-B4), and with the resulting unit estimator gain established in Lemma 1.
- Step 4 of Algorithm 1 solves exactly the convex program of Eq. (4); existence, uniqueness, and non-expansiveness of this per-step map are guaranteed by Theorem 2, and the empirical K_p -insensitivity of Table II is explained by the saturation mechanism of Proposition 1, not treated as an unexplained coincidence.
- Step 5's quantization satisfies the deterministic error bound of Proposition 2, which enters the practical-stability statement of Theorem 3 as a bounded perturbation.
- Step 7's advancement of the SUMO physical layer is the L1 component whose non-divergence is verified empirically in Section V, consistent with the explicit scope limitation stated in Theorem 2 and Theorem 3: no closed-form contraction of the physical traffic system is claimed, only boundedness/Lipschitz continuity of the allocator (L3) and a deterministic bound on quantization error (L4).

Every algorithmic parameter thus has a direct, honestly scoped mapping to a theoretical quantity: the sliding window and its normalization provide the statistical foundation (L2), the convex program provides the actuation with a genuine role for K_p (L3), the quantization bound accounts for integer execution (L4), and the physical loading model (L1) is stated explicitly rather than left implicit, with its stability verified empirically rather than claimed analytically.

V. EXPERIMENT

A. Experiment Setup

a) *Network and Traffic Loading Model*: Experiments are conducted on a cropped subregion of the Philadelphia street network. The full pre-crop network comprises 13,389 nodes and 40,003 edges; applying a geographic bounding-box crop to node coordinates at `scale_ratio = 0.2` yields a cropped subnetwork of 4,186 edges spanning 1,474 nodes. The simulation is run end-to-end in SUMO under the microscopic car-following traffic loading model (`TRAFFIC_LOADING_MODEL = SUMO_MICROSCOPIC_CARFOLLOWING`); vehicle motion, link-capacity constraints, queuing, spillback, and travel time are all produced endogenously by SUMO's microscopic dynamics rather than by an abstract loading model, in contrast to smaller benchmark networks such as Sioux Falls. Demand is drawn from a fixed pool of up to `TARGET_TRIP_COUNT = 1,000,000` origin–destination trips; within the cropped region and the 1800 s departure window, the eligible trip pool itself totals 741,449 trips (below the 1,000,000 cap, so all eligible trips

are retained rather than subsampled), spanning 175 distinct origin junctions, 188 distinct destination junctions, and 27,546 unique origin–destination junction pairs. Candidate paths are precomputed via k -shortest paths ($k = 3$) for every OD pair prior to cropping (mean path length ≈ 80 edges over the full network); a candidate path survives the crop only if all of its constituent edges lie within the cropped subnetwork, so a surviving OD pair retains between 1 and 3 valid candidate paths after cropping.

b) Simulation Protocol.: Each run follows a three-phase protocol: a 300 s warm-up period (WARMUP_DURATION) during which the network is loaded but not scored, a 1800 s scored test window (SIM_DURATION), and a 600 s clearance period (CLEARANCE_DURATION) appended after the test window closes. This mitigates empty-network initialization bias and truncated-horizon effects that can arise in shorter simulation windows. Because trip departures are sampled at city scale and the scored window is comparatively short relative to typical city-scale trip lengths, a non-trivial share of vehicles has not completed its trip by the end of the scored window; we report this directly rather than restricting statistics to completed trips only, since conditioning on completion alone would bias congestion and travel-time statistics toward the fastest-clearing vehicles. The incomplete-vehicle ratio is reported per configuration in Table II.

c) Congestion Threshold Calibration.: Following the calibration/validation/test separation required to avoid post-hoc threshold tuning, occupancy statistics are first collected from three dedicated calibration seeds (CALIBRATION_SEEDS = {100, 101, 102}), disjoint from the three validation seeds (VALIDATION_SEEDS = {200, 201, 202}) and the held-out test seed (TEST_SEEDS = {42}). For a quantile $q \in [0, 1]$, the threshold is the empirical quantile of the non-zero calibration occupancy samples,

$$\theta(q) = Q_q(\{o_{t,e} : o_{t,e} > 0\}), \quad (11)$$

rather than a linear min–max rescaling of the form $\theta(q) = \text{occ}_{\min} + q(\text{occ}_{\max} - \text{occ}_{\min})$, which does not correspond to a statistical quantile. The candidate quantile q is selected on the validation seeds and then frozen for evaluation on the held-out test seed, so no threshold is tuned against the numbers being reported.

d) Baselines and Control/Prediction Separation.: We compare the proposed controller (*Original Endogenous*) against five baselines, explicitly partitioned by operational mode to avoid conflating prediction accuracy with control efficacy: (i) *D³STN-Enhanced*, a spatiotemporal graph neural network whose congestion-risk output is fed directly into the same feedback-actuation logic as the proposed method, making it the only learned baseline with an active actuator; (ii) *Marginal-Cost Routing* and (iii) *MPC Routing*, two classical closed-loop control baselines with their own actuation logic; and (iv) *STGCN*, (v) *DCRNN*, and (vi) *ST-ResNet*, three open-loop occupancy predictors whose `allocate_vehicles()` call is a no-op — they never execute a routing adjustment. Consistent with this split, results are reported in two structurally separate tables: a **Control Performance Table** (Original Endogenous, D³STN-Enhanced, Marginal-Cost Routing,

MPC Routing) reporting travel time, congestion, and recovery metrics for the four closed-loop actuators, and a **Prediction Accuracy Table** (STGCN, DCRNN, ST-ResNet) reporting MAE, RMSE, and Top-10% congestion-overlap for the three open-loop predictors. No travel-time or control-performance figures are reported for the predictors, since they do not actuate any route diversion and such a comparison would not be meaningful.

e) Metric Scope.: Precision and recall are not reported as headline controller performance metrics for the closed-loop controllers (Original Endogenous, D³STN-Enhanced, Marginal-Cost Routing, MPC Routing). These are detection-quality metrics for an open-loop forecaster; for a closed-loop actuator whose routing decisions are themselves conditioned on the detection signal, precision/recall of that internal signal do not characterize controller performance. Detection precision/recall is retained only internally, as the objective used to select the frozen congestion threshold in the calibration/validation/test procedure above (Eq. 11); it is not reported as a per-algorithm result.

B. Results: Control Performance and Prediction Accuracy

C. Explanation

a) Travel Time and Congestion Duration.: The proposed method achieves the lowest average travel time among all four closed-loop controllers (980.2 s), marginally ahead of D³STN-Enhanced (983.6 s, +0.35%), MPC Routing (982.4 s, +0.22%), and Marginal-Cost Routing (992.6 s, +1.27%). It simultaneously achieves the lowest cumulative congestion time (329,159 edge-s), again ahead of MPC Routing (329,620), D³STN-Enhanced (334,797), and Marginal-Cost Routing (348,257). On this Philadelphia network, the proposed method does not trade travel time for congestion mitigation against the other closed-loop baselines — it is simultaneously best or near-best on both axes — while requiring no offline pre-training and no adjacency matrix, unlike D³STN-Enhanced.

b) Recovery and Stability.: D³STN-Enhanced attains the fastest average recovery time (241.4 s), narrowly ahead of MPC Routing (242.5 s) and the proposed method (265.3 s), while Marginal-Cost Routing is slowest to recover (307.1 s). Average congestion depth and occupancy standard deviation are effectively tied across all four controllers (0.0007–0.0008 and 0.0011 respectively), indicating that, at this operating point, the controllers differ more in *when* and *how long* congestion persists than in its peak severity or network-wide dispersion.

c) Computational Efficiency.: MPC Routing has the lowest per-step latency (15.20 ms), with the proposed method close behind (23.14 ms) and Marginal-Cost Routing comparable (23.53 ms); D³STN-Enhanced is substantially more expensive (87.54 ms/step, 3.8 $\times$ the proposed method) due to its GNN forward pass. Among the three open-loop predictors, ST-ResNet is cheapest (22.78 ms/step) and DCRNN most expensive (109.83 ms/step). Only the proposed method and the two classical control baselines (Marginal-Cost, MPC) require neither pre-training nor an adjacency matrix.

TABLE I
CONTROL PERFORMANCE TABLE (CONGESTION THRESHOLD = 0.0047)

Metric	Original Endogenous (Ours)	D ³ STN-Enhanced	Marginal-Cost Routing	MPC Routing
Control condition	Closed-loop feedback control			
Avg Travel Time (s)	980.2	983.6	992.6	982.4
Total Congestion Time (edge-s)	329159	334797	348257	329620
Avg Congestion Depth	0.0007	0.0007	0.0008	0.0007
Avg Congestion Duration (s)	337.9	336.1	354.3	326.4
Avg Recovery Time (s)	265.3	241.4	307.1	242.5
Occupancy Std Dev	0.0011	0.0011	0.0011	0.0011
Computation Latency (CPU, ms/step)	23.14	87.54	23.53	15.20
Requires Pre-training?	No	Yes	No	No
Requires Adjacency Matrix?	No	Yes	No	No

Note: All four methods are closed-loop actuators; none of the open-loop predictors (STGCN/DCRNN/ST-ResNet) are included here (see Table II). Bold marks the best value in each row among the four control methods, with ties bolded jointly; the Occupancy Std Dev row is left unbolded since all four methods report an identical value.

TABLE II
PREDICTION ACCURACY TABLE (CONGESTION THRESHOLD = 0.0047)

Metric	STGCN Baseline	DCRNN Baseline	ST-ResNet Baseline
MAE	0.0007	0.0015	0.0032
RMSE	0.0015	0.0021	0.0033
Top-10% Overlap	6.31%	5.93%	8.85%
Computation Latency (CPU, ms/step)	65.50	109.83	22.78

Note: These three strategies are open-loop occupancy predictors; `allocate_vehicles()` returns a no-op assignment, so no travel-time or control-performance columns are reported for them.

d) Prediction Accuracy Is Disjoint From Control Efficacy. Table II reports forecasting accuracy for the three open-loop predictors purely for completeness with the traffic-forecasting literature; none of the three ever executes a routing decision. ST-ResNet attains the best Top-10% congestion-overlap rate (8.85%) among the predictors, yet by construction has no effect on travel time or congestion duration, since `allocate_vehicles()` is a no-op for all three. This reinforces the paper’s central claim: forecasting accuracy and control efficacy are measured on structurally disjoint axes, and a model’s overlap/MAE/RMSE score says nothing about whether it can be deployed as a controller without a dedicated actuation mechanism such as the one embedded in the proposed method or in D³STN-Enhanced.

D. Ablation Experiment (K_p)

E. Explanation

a) Detection-Only Baseline vs. Closed-Loop Actuation. On the Philadelphia network the detection-only baseline ($K_p = 0.00$) is not competitive on any axis: it has the *highest* average trip time (1125.9 s, +5.2% to +5.8% over the closed-loop settings), the highest cumulative congestion time (39,728 edge-s), the longest average lead time before a valid diversion detection (529.1 s), and the lowest diversion trigger ratio (0.388). Enabling closed-loop actuation at any tested $K_p > 0$ therefore improves travel time and congestion mitigation simultaneously, with no user-equilibrium-vs-resilience trade-off to negotiate: activating the feedback loop is unambiguously better on this network.

b) Graded Response Below Saturation, Collapse Above It. Under the primary operating point ($\gamma_{\max} = 0.7$), the

three closed-loop gains produce distinct, non-degenerate outcomes: total congestion time decreases monotonically from $K_p = 0.01$ to $K_p = 0.02$ with $K_p = 0.05$ essentially matching $K_p = 0.02$ (37,793); average trip time is lowest at $K_p = 0.02$ (1064.3 s), 0.08% higher at $K_p = 0.01$, and 0.54% higher at $K_p = 0.05$. This confirms that K_p exerts a graded, controllable influence on the allocation, in contrast to the flat, gain-independent behavior that would occur under an unconstrained hard-clip configuration with $\gamma_{\max} = 1.0$. Under the tighter cap ($\gamma_{\max} = 0.4$), however, all three gains $K_p \in \{0.01, 0.02, 0.05\}$ converge to *identical* values on every non-latency metric — avg trip time, congestion time, recovery time, lead time, and trigger ratio are all numerically equal across the three settings. This is the empirical signature of the KKT complementary-slackness saturation mechanism (Proposition 1): once the box constraint $\gamma_{\max}$ is the binding constraint for a sufficiently wide range of paths, further increases in the penalty coefficient K_p no longer change the projected solution, so any K_p above the saturation point is equivalent. We report both regimes rather than only the more favorable one, since the saturation behavior under $\gamma_{\max} = 0.4$ is itself a direct, testable prediction of the theory rather than an artifact to be hidden.

c) Choice of Default K_p . We adopt $K_p = 0.02$ as the default operating point: it attains the best or joint-best value on total congestion time, average trip time, and average lead time under the primary sweep, while remaining inside the graded (non-saturated) regime at $\gamma_{\max} = 0.7$, which is the setting used for the Control Performance Table (Table I).

d) Computation Latency Under Actuation. Enabling closed-loop actuation increases per-step latency relative to the detection-only baseline by roughly 0.39–0.40 ms under the primary sweep (0.76 ms $\rightarrow$ 1.14–1.15 ms) but only 0.24–

TABLE III
 K_p ABLATION STUDY (ORIGINAL ENDOGENOUS ONLY; CONGESTION THRESHOLD = 0.0045)

Metric	Free-flow weighted baseline ($K_p = 0.00$)	$K_p = 0.01$	$K_p = 0.02$	$K_p = 0.05$
Primary Sweep ($\gamma_{\max} = 0.7$)				
Diversion mode	Detection-only (open-loop)			Closed-loop actuation
Avg Trip Time (s)	1125.9	1065.2	1064.3	1070.0
Total Congestion Time (edge·s)	39728	38213	37703	37793
Avg Recovery Time (s)	251.2	194.5	198.5	276.8
Avg Lead Time (s)	529.1	438.7	438.7	476.1
Diversion Trigger Ratio	0.388	0.444	0.446	0.449
Incomplete Vehicle Ratio	0.953	0.952	0.952	0.951
Computation Latency (CPU, ms/step)	0.756	1.143	1.154	1.142
Hard-Saturation Comparison ($\gamma_{\max} = 0.4$)				
Diversion mode	Detection-only (open-loop)			Closed-loop actuation
Avg Trip Time (s)	1125.9	1070.0	1070.0	1070.0
Total Congestion Time (edge·s)	39728	37793	37793	37793
Avg Recovery Time (s)	251.2	276.8	276.8	276.8
Avg Lead Time (s)	529.1	476.1	476.1	476.1
Diversion Trigger Ratio	0.388	0.449	0.449	0.449
Incomplete Vehicle Ratio	0.953	0.951	0.951	0.951
Computation Latency (CPU, ms/step)	0.941	1.192	1.180	1.188

Note: $K_p = 0$ is labeled “Free-flow weighted baseline” (never “User Equilibrium”), and reported as *Detection-only* rather than open-loop no-control, since it still runs congestion detection but performs no actuation. All non-latency figures under $\gamma_{\max} = 0.4$ for $K_p \in \{0.01, 0.02, 0.05\}$ are numerically identical, evidencing box-constraint saturation (Proposition 1); bold marks the best value among the four K_p settings within each sweep.

0.25 ms under hard saturation (0.94 ms $\rightarrow$ 1.18–1.19 ms); the smaller overhead under $\gamma_{\max} = 0.4$ is consistent with fewer paths remaining active in the water-filling iteration once most are already pinned at their capacity bound. This remains three orders of magnitude below the 1 s control interval required for real-time operation, and is an order of magnitude cheaper than the D³STN-Enhanced baseline’s 87.54 ms/step reported in Table I.

e) Incomplete Vehicle Ratio.: Across all eight configurations in Table III, roughly 95% of vehicles that departed during the 1800 s scored window have not reached their destination by the end of that window; this ratio is essentially unaffected by K_p or $\gamma_{\max}$ (0.951–0.953 throughout). We report this figure directly, consistent with the requirement to disclose the fraction of incomplete trips rather than computing travel-time statistics only over completed vehicles, which would bias the reported averages toward the fastest-clearing trips. The near-constant incomplete ratio across all eight configurations also indicates that the differences reported above in trip time, congestion time, and lead time are not an artifact of differential sample selection between configurations.

f) Threshold Discrepancy Note.: The Control Performance Table (Table I) uses threshold 0.0047, while the K_p ablation study (Table III) uses threshold 0.0045. Both are outputs of the same calibration procedure (Eq. 11) applied to independently drawn calibration-seed samples in the two experiment scripts; the small difference reflects sampling variance in the calibration set rather than a change in methodology, and does not materially affect either table’s internal comparisons, which are computed within a fixed threshold. The $T_w \times \alpha$ sensitivity sweep (Table IV) instead uses a fixed, uncalibrated threshold of 0.005 rather than either calibrated value above; this was a deliberate simplification to isolate the estimator hyperparameters’ effect from threshold-calibration

variance, and comparisons within Table IV should not be read against the 0.0047/0.0045 figures in Tables I–III.

F. Sensitivity to Estimator Hyperparameters (T_w, α)

a) Robustness of Travel Time to (T_w, α).: Average trip time varies by less than 1% across all 16 configurations (977.7–985.9 s), with no monotonic trend in either T_w or α . This indicates the controller’s primary performance outcome is not brittle to the choice of sliding-window length or decay factor, supporting the framing of $T_w = 60$ s as an engineering default rather than a provably optimal value.

b) Trigger Ratio Increases Monotonically With α .: Within every fixed T_w , the diversion trigger ratio increases monotonically with α (e.g., $T_w = 15$: 0.385 $\rightarrow$ 0.407 $\rightarrow$ 0.426 $\rightarrow$ 0.435; $T_w = 120$: 0.416 $\rightarrow$ 0.447 $\rightarrow$ 0.475 $\rightarrow$ 0.490). This is consistent with the estimator’s construction: a larger α retains more weight on older observations, smoothing the risk estimate and making triggered detections more persistent once congestion is under way. The trigger ratio also rises mildly with T_w at fixed $\alpha = 0.99$ (0.435 $\rightarrow$ 0.454 $\rightarrow$ 0.482 $\rightarrow$ 0.490), as longer windows accumulate more corroborating evidence before a detection is confirmed.

c) Latency Grows With Window Length, With Caveats.: Averaged within each T_w group, per-step computation latency rises from 10.82 ms ($T_w = 15$) and 10.93 ms ($T_w = 30$) to 12.98 ms ($T_w = 60$) and 14.89 ms ($T_w = 120$), consistent with the $O(T_w)$ cost of maintaining a longer sliding-window buffer. The $T_w = 15$ and $T_w = 30$ groups are not reliably distinguishable from one another (means differ by 0.11 ms, well within their internal spread of 9.2–13.7 ms at $T_w = 30$), so we report this as a group-level trend separating $\{15, 30\}$ from $\{60, 120\}$ rather than a strictly monotonic per-configuration relationship. Even at $T_w = 120$, latency remains roughly two orders of magnitude below the 1 s control interval.

TABLE IV
SENSITIVITY SWEEP: SLIDING-WINDOW LENGTH $T_w \times$ DECAY FACTOR α (ORIGINAL ENDOGENOUS, $\gamma_{\max} = 0.7$, $K_p = 0.02$, CONGESTION THRESHOLD = 0.005)

T_w (s)	α	Avg Trip Time (s)	Total Congestion Time (edge-s)	Diversion Trigger Ratio	Latency (CPU, ms/step)
15	0.85	977.8	234438	0.385	9.40
15	0.90	978.5	232049	0.407	11.49
15	0.95	979.1	231536	0.426	10.54
15	0.99	981.2	231223	0.435	11.87
30	0.85	977.7	234176	0.408	13.70
30	0.90	981.0	233976	0.432	9.21
30	0.95	978.1	227463	0.452	9.83
30	0.99	981.7	235277	0.454	11.00
60	0.85	981.2	230644	0.416	12.03
60	0.90	983.8	232992	0.442	12.76
60	0.95	980.8	234887	0.471	13.33
60	0.99	977.8	236112	0.482	13.79
120	0.85	985.9	235904	0.416	14.24
120	0.90	983.6	240058	0.447	14.71
120	0.95	980.9	230524	0.475	15.23
120	0.99	980.8	228406	0.490	15.39

Note: All 16 combinations use the fixed operating point $\gamma_{\max} = 0.7$, $K_p = 0.02$ established in Table III. Bold marks the best value in each column. This sweep reports control-outcome sensitivity (travel time, congestion time, trigger ratio, latency) rather than the estimator's own bias-variance decomposition $\text{MSE}(T_w) = \text{Bias}^2 + \text{Variance}$; the latter would require ground-truth arrival-rate labels and is left for future work.

VI. CONCLUSION

This paper has presented an endogenous, event-triggered congestion control framework for urban road networks that requires no external reference signal, no plant model, and no offline training. Route reallocation is posed as a constrained convex program in which the feedback gain K_p enters as a genuine penalty coefficient. We established, for this per-step allocation map, existence, uniqueness, and non-expansiveness with respect to the risk estimate, together with a deterministic quantization-error bound for integer vehicle assignment and a practical, bounded-perturbation stability statement for the resulting closed loop. These guarantees are deliberately scoped to the allocator's own input-output behavior; the physical traffic layer is realized by SUMO's microscopic simulation, has no closed-form model, and its non-divergence is verified empirically rather than claimed as an analytic global stability result.

Experimental evaluation on a large-scale SUMO simulation of a cropped Philadelphia network supports these design choices. Enabling closed-loop actuation reduces both average travel time and cumulative congestion time simultaneously relative to a detection-only baseline, with no trade-off between the two observed at this operating point. The proposed controller further attains travel time and congestion performance comparable to or better than a GNN-enhanced baseline (D³STN-Enhanced) and two classical control baselines (Marginal-Cost and MPC routing), while requiring roughly one-quarter of the GNN baseline's per-step computation time and no pre-training or adjacency matrix.

The ablation over K_p provides direct empirical support for the corrected theory: below the network's capacity-cap saturation point, increasing K_p produces distinct, non-degenerate improvements in congestion time and travel time; above it, all tested gains converge to numerically identical outcomes. This is the behavior predicted by the KKT complementary-

slackness argument underlying the allocator's convex program, rather than an unexplained empirical coincidence.

Several limitations remain. First, statistical significance testing across multiple independent simulation seeds—using seeds, rather than individual vehicles or edge-timesteps, as the sampling unit—has not yet been completed for this network and is necessary before the reported differences can be characterized as statistically robust rather than single-seed observations. Second, a sensitivity sweep over the estimator's window length and decay factor indicates that average travel time is not brittle to these choices, but this is a control-outcome sensitivity rather than a formal bias-variance-optimality result, and the 60-second window remains an engineering default rather than a provably optimal one. Third, the current framework assumes a fixed OD demand pool and a static candidate-path set; time-varying demand would require online path-set adaptation. Fourth, communication delay, packet loss, and sensor noise are examined only in separate perturbation experiments rather than the main results, and practical deployment would need to account for them directly. Finally, validation has so far been carried out on a single cropped urban network; evaluation on additional, topologically distinct networks would further support generalizability.

In summary, this paper replaces an informally stated control law with an explicit constrained-optimization formulation in which the feedback gain plays a genuine, analytically accounted-for role, and evaluates the resulting controller on a large-scale microscopic traffic simulation rather than an abstract loading model. The results indicate that endogenous, model-free feedback from sliding-window vehicle counts can mitigate congestion without the plant models, offline training, or graph-topology dependence required by learning-based alternatives, while remaining computationally lightweight; broader validation across networks and a completed multi-seed statistical analysis are the next steps toward establishing this

more generally.

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REFERENCES

- [1] L. Breiman, "The Poisson Tendency in Traffic Distribution," *The Annals of Mathematical Statistics*, vol. 34, no. 1, pp. 308 – 311, 1963. [Online]. Available: <https://doi.org/10.1214/aoms/1177704267>
- [2] W. J. Gordon and G. F. Newell, "Equilibrium analysis of a stochastic model of traffic flow," *Mathematical Proceedings of the Cambridge Philosophical Society*, vol. 60, no. 2, pp. 227–236, Apr. 1964. [Online]. Available: https://www.cambridge.org/core/product/identifier/S0305004100037695/type/journal_article
- [3] A. Rényi, "On two mathematical models of the traffic on a divided highway," *Journal of Applied Probability*, vol. 1, pp. 311 – 320, 1964. [Online]. Available: <https://api.semanticscholar.org/CorpusID:222416666>
- [4] P. I. Richards, "Shock waves on the highway," *Operations Research*, vol. 4, no. 1, pp. 42–51, 1956. [Online]. Available: <http://www.jstor.org/stable/167515>
- [5] D.-w. Xu, Y.-d. Wang, L.-m. Jia, Y. Qin, and H.-h. Dong, "Real-time road traffic state prediction based on ARIMA and Kalman filter," *Frontiers of Information Technology & Electronic Engineering*, vol. 18, no. 2, pp. 287–302, Feb. 2017. [Online]. Available: <http://link.springer.com/10.1631/FITEE.1500381>
- [6] V. I. Kontopoulou, A. D. Panagopoulos, I. Kakkos, and G. K. Matsopoulos, "A Review of ARIMA vs. Machine Learning Approaches for Time Series Forecasting in Data Driven Networks," *Future Internet*, vol. 15, no. 8, p. 255, Jul. 2023. [Online]. Available: <https://www.mdpi.com/1999-5903/15/8/255>
- [7] Y. Li, R. Yu, C. Shahabi, and Y. Liu, "Diffusion Convolutional Recurrent Neural Network: Data-Driven Traffic Forecasting," arXiv, Feb. 2018, arXiv:1707.01926 [cs.LG]. [Online]. Available: <http://arxiv.org/abs/1707.01926>
- [8] J. J. Yu, "Scalable and Sustainable Graph-Based Traffic Prediction With Adaptive Deep Learning," *IEEE Transactions on Knowledge and Data Engineering*, vol. 36, no. 11, pp. 6949–6961, Nov. 2024. [Online]. Available: <https://ieeexplore.ieee.org/document/10571386/>
- [9] S. Singha, D. Agarwal, and N. Samal, "A Review of Various Prediction Techniques in the Context of Traffic Congestion Prediction," in *2024 2nd International Conference on Signal Processing, Communication, Power and Embedded System (SCOPES)*. Paralakhemundi Campus, Centurion University of Technology and Management, Odisha., India: IEEE, Dec. 2024, pp. 1–6. [Online]. Available: <https://ieeexplore.ieee.org/document/10990870/>
- [10] J. Zhang, Y. Zheng, and D. Qi, "Deep Spatio-Temporal Residual Networks for Citywide Crowd Flows Prediction," arXiv, Jan. 2017, arXiv:1610.00081 [cs.AI]. [Online]. Available: <http://arxiv.org/abs/1610.00081>
- [11] Q. Yuan, J. Wang, Y. Han, Z. Liu, and W. Liu, "DAGCAN: Decoupled Adaptive Graph Convolution Attention Network for Traffic Forecasting," *IEEE Transactions on Intelligent Transportation Systems*, vol. 26, no. 3, pp. 3513–3526, Mar. 2025. [Online]. Available: <https://ieeexplore.ieee.org/document/10874891/>
- [12] Y. Ba, S. Ji, S. Pan, S. He, Y. Ji, and D. Guo, "Urban Road Traffic Prediction With Dynamic Spatial-Temporal Graph Convolutional Network Based on Attention Mechanism," *IEEE Sensors Journal*, vol. 25, no. 14, pp. 27187–27199, Jul. 2025. [Online]. Available: <https://ieeexplore.ieee.org/document/11031099/>
- [13] J. Fan, F. Zhu, W. Weng, X. Zhang, H. Jiang, H. Tian, and H. Wu, "Dynamic Modeling and Analysis of Bi-Directional Traffic Flows Through a Deep Spatio-Temporal Graph Neural Network," *IEEE Transactions on Big Data*, vol. 11, no. 6, pp. 3016–3028, Dec. 2025. [Online]. Available: <https://ieeexplore.ieee.org/document/10994373/>
- [14] G. Fan, A. Q. M. Sabri, S. S. A. Rahman, and L. Pan, "DynaKey-GNN: An efficient dynamic key-node multi-graph neural network for spatio-temporal traffic flow forecasting," *Engineering Applications of Artificial Intelligence*, vol. 159, p. 111757, Nov. 2025. [Online]. Available: <https://linkinghub.elsevier.com/retrieve/pii/S0952197625017592>
- [15] S. Jiang, Q. Liu, and X. Dai, "Ricci-Curvature-Enhanced Temporal Graph Convolutional Networks for Traffic Speed Forecasting," *IEEE Access*, vol. 14, pp. 20414–20426, 2026. [Online]. Available: <https://ieeexplore.ieee.org/document/11373167/>
- [16] S. Van Tran and V. H. Duong, "Transformer-Based Traffic Flow Forecasting: A Survey and Experimental Comparison of Time-Series Analysis," in *2025 RIVF International Conference on Computing and Communication Technologies (RIVF)*. Ho Chi Minh City, Vietnam: IEEE, Dec. 2025, pp. 1303–1308. [Online]. Available: <https://ieeexplore.ieee.org/document/11365187/>
- [17] W. Zhu, X. Zhang, C. Liu, and Y. Sun, "D³STN: Dynamic Delay Differential Equation Spatiotemporal Network for Traffic Flow Forecasting," *IEEE Transactions on Intelligent Transportation Systems*, vol. 25, no. 11, pp. 18093–18106, Nov. 2024. [Online]. Available: <https://ieeexplore.ieee.org/document/10642992/>
- [18] C. Yu, X. Xie, Y. Huang, and C. Qiu, "FUSE-Traffic: Fusion of Unstructured and Structured Data for Event-aware Traffic Forecasting," in *Proceedings of the 33rd ACM International Conference on Advances in Geographic Information Systems*, Nov. 2025, pp. 620–630, arXiv:2510.16053 [cs.LG]. [Online]. Available: <http://arxiv.org/abs/2510.16053>
- [19] S. Shafiei, Q. Zhu, A. K. Qin, S. A. Mihaita, H. Grzybowski, and H. Dia, "Road traffic attributes prediction using deep learning hybridization by the traffic fundamental diagram," *Journal of Intelligent Transportation Systems*, pp. 1–16, Oct. 2025. [Online]. Available: <https://www.tandfonline.com/doi/full/10.1080/15472450.2025.2577735>
- [20] Y. Xu and Y. Li, "A Real-Time Urban Traffic Congestion Prediction Framework Based on Dynamic Risk Field and Multi-Source Data Fusion," *IEEE Access*, vol. 13, pp. 172471–172495, 2025. [Online]. Available: <https://ieeexplore.ieee.org/document/11159209/>
- [21] W. Sun, H. Cui, and M. Zhu, "Traffic Congestion Analysis and Probability Estimation Based on Stochastic Characteristics of Traffic Arrival," *Promet - Traffic & Transportation*, vol. 37, no. 3, pp. 585–597, Jun. 2025. [Online]. Available: <https://traffic2.fpz.hr/index.php/PROMTT/article/view/965>
- [22] X. Wang, Y. Tang, N. Espitia, and N. Bekiaris-Liberis, "Event-Triggered Control of Freeway Traffic Flow with Connected and Automated Vehicles," *IFAC-PapersOnLine*, vol. 59, no. 8, pp. 31–36, 2025. [Online]. Available: <https://linkinghub.elsevier.com/retrieve/pii/S2405896325006469>

APPENDIX

Metric Definitions (Tables I–IV)

$k = 1, \dots, K$ indexes post-warm-up recorded evaluation steps ($W_{\text{warm}} = 300$ s); $\tilde{o}_e(k)$ is the length- T_w causal sliding-window mean of edge e 's raw SUMO occupancy; θ^{real} is the scalar real-congestion threshold; $r_e(k)$ is the predicted risk and θ_e the edge-specific detection threshold; $\mathcal{E}$ is the edge set.

a) *Avg Travel Time / Avg Trip Time (s)*: Let $\mathcal{V}^{\text{comp}}$ be the set of vehicles with a completed `tripinfo` record and $d(v)$ the SUMO-reported trip duration of vehicle $v \in \mathcal{V}^{\text{comp}}$. Then

$$\text{avg_trip_time} = \begin{cases} \frac{1}{|\mathcal{V}^{\text{comp}}|} \sum_{v \in \mathcal{V}^{\text{comp}}} d(v), & |\mathcal{V}^{\text{comp}}| > 0 \\ 0, & |\mathcal{V}^{\text{comp}}| = 0 \end{cases} \quad (12)$$

This average is taken only over vehicles that produced a completion record; incomplete vehicles are excluded here and accounted for separately via `incomplete_ratio` below.

b) *Total Congestion Time (edge-s)*:

$$\text{total_congestion_time} = \sum_{e \in \mathcal{E}} \sum_{k=1}^K \mathbb{I}[\tilde{o}_e(k) \geq \theta^{\text{real}}] \quad (13)$$

Total edge-seconds spent at or above the global real-congestion threshold, summed over all edges and all recorded steps.

c) *Avg Congestion Depth.*:

$$S = \{(e, k) : \tilde{o}_e(k) \geq \theta^{\text{real}}\} \quad (14)$$

$$\text{avg_congestion_depth} = \begin{cases} \frac{1}{|S|} \sum_{(e,k) \in S} (\tilde{o}_e(k) - \theta^{\text{real}}), \\ 0, \end{cases} \quad (15)$$

Mean overshoot above threshold, averaged only over the edge-second samples that are actually congested.

d) *Avg Congestion Duration (s).*: For each edge e , let $\{[s_i^e, f_i^e]\}_i$ be the maximal contiguous runs of steps for which $\mathbb{I}[\tilde{o}_e(k) \geq \theta^{\text{real}}] = 1$ (identified via zero-padded rising/falling edges of the indicator sequence). Then

$$\text{real_avg_congestion_duration} = \begin{cases} \frac{1}{N_{\text{ep}}} \sum_e \sum_i (f_i^e - s_i^e), & N_{\text{ep}} > 0 \\ 0, & N_{\text{ep}} = 0 \end{cases} \quad (16)$$

where N_{ep} is the total number of congestion episodes pooled across all edges. This is an event-level average (per episode), not a per-edge average.

e) *Avg Recovery Time (s).*: For each edge e , let $k_e^* = \arg \max_k \tilde{o}_e(k)$ be the time of its single episode-wide peak smoothed occupancy, and define the recovery lag

$$\delta_e = \min \{ \delta \geq 0 : \tilde{o}_e(k_e^* + \delta) < \theta^{\text{real}} \} \quad (17)$$

whenever $\tilde{o}_e(k_e^*) \geq \theta^{\text{real}}$ and such a δ exists within the recorded horizon (edges without a qualifying post-peak recovery are excluded). Then, with $\mathcal{E}^{\text{rec}}$ the set of qualifying edges,

$$\text{avg_recovery_time} = \begin{cases} \frac{1}{|\mathcal{E}^{\text{rec}}|} \sum_{e \in \mathcal{E}^{\text{rec}}} \delta_e, & \mathcal{E}^{\text{rec}} \neq \emptyset \\ 0, & \mathcal{E}^{\text{rec}} = \emptyset \end{cases} \quad (18)$$

Only each edge's single global peak is examined, not every congestion episode on that edge.

f) *Occupancy Std Dev.*:

$$\text{real_occupancy_std} = \text{std}_{e \in \mathcal{E}} \left(\frac{1}{K} \sum_{k=1}^K \tilde{o}_e(k) \right) \quad (19)$$

Time-average each edge first, then take the standard deviation across edges; this measures spatial heterogeneity of average congestion, not temporal volatility.

g) *Avg Lead Time (s) / Diversion Trigger Ratio.*: Let $k_e^{\text{pred}} = \min\{k : r_e(k) \geq \theta_e\}$ and $k_e^{\text{real}} = \min\{k : \tilde{o}_e(k) \geq \theta^{\text{real}}\}$ (each possibly undefined), $\mathcal{E}^{\text{cong}} = \{e : k_e^{\text{real}} \text{ exists}\}$, and $L_{\text{max}} = 1800$ s. Define

$$\tau_e^{\text{lead}} = \begin{cases} k_e^{\text{real}} - k_e^{\text{pred}}, & e \in \mathcal{E}^{\text{cong}}, k_e^{\text{pred}} \text{ exists,} \\ 0 \leq k_e^{\text{real}} - k_e^{\text{pred}} \leq L_{\text{max}} \\ \text{NaN,} & \text{otherwise} \end{cases} \quad (20)$$

$$\text{avg_lead_second} = \text{nanmean}_e(\tau_e^{\text{lead}}) \quad (0 \text{ if all NaN}) \quad (21)$$

Separately, with $b_e(k) = \mathbb{I}[r_e(k) \geq \theta_e]$ the predicted-trigger mask,

$$\text{business_trigger_ratio} = \frac{1}{K \cdot |\mathcal{E}|} \sum_{e \in \mathcal{E}} \sum_{k=1}^K b_e(k) \quad (22)$$

The pooled fraction of all (edge, step) pairs with an active predicted trigger.

h) *Incomplete Vehicle Ratio.*: Let $\mathcal{V}^{\text{dep}}$ be all vehicles that ever departed (`traci.simulation.getDepartedIDList()`, accumulated over the run) and $\mathcal{V}^{\text{comp}}$ the completed set defined above. Then

$$\text{no_complete_ratio} = \begin{cases} \frac{|\mathcal{V}^{\text{dep}} \setminus \mathcal{V}^{\text{comp}}|}{|\mathcal{V}^{\text{dep}}|}, & \mathcal{V}^{\text{dep}} \neq \emptyset \\ 0, & \mathcal{V}^{\text{dep}} = \emptyset \end{cases} \quad (23)$$

Fraction of departed vehicles with no completion record by the end of the loading-plus-clearance window.

i) *Computation Latency (CPU, ms/step).*: Let T_{algo} be the cumulative wall-clock time spent only inside the tracker-update, risk-computation, and allocation calls (excluding SUMO/TraCI stepping itself), and T_{sim} the nominal scored duration in seconds. Then

$$\text{avg_step_ms} = \frac{T_{\text{algo}}}{T_{\text{sim}}} \times 1000 \quad (24)$$

Because the simulation loop actually continues for $T_{\text{sim}} + \text{CLEARANCE_DURATION}$ seconds (up to 600s more) unless the network clears earlier, and algorithmic time accrued during this clearance tail is still included in T_{algo} 's numerator while the denominator remains T_{sim} only, this metric mildly overstates the true per-elapsed-second rate.

j) *MAE / RMSE (Prediction Accuracy Table).*: With $o_e(k)$ the raw, unsmoothed SUMO occupancy,

$$\text{raw_mae} = \frac{1}{K|\mathcal{E}|} \sum_{e \in \mathcal{E}} \sum_{k=1}^K |r_e(k) - o_e(k)| \quad (25)$$

$$\text{raw_rmse} = \sqrt{\frac{1}{K|\mathcal{E}|} \sum_{e \in \mathcal{E}} \sum_{k=1}^K (r_e(k) - o_e(k))^2} \quad (26)$$

Both compare predicted risk against raw occupancy, not the smoothed $\tilde{o}_e(k)$ used in the congestion/trigger metrics above.

k) *Top-10% Overlap.*: Let $\rho = 0.1$ and $\kappa = \lfloor \rho \cdot K|\mathcal{E}| \rfloor$. Let $\text{TopK}_{\kappa}(\cdot)$ select the κ largest entries of a flattened (edge, step) array. Then

$$\text{top_hit_ratio} = \frac{|\text{TopK}_{\kappa}(\{r_e(k)\}) \cap \text{TopK}_{\kappa}(\{\tilde{o}_e(k)\})|}{\kappa} \quad (27)$$

Overlap between the globally highest-risk and globally highest-occupancy (edge, time) cells, with no temporal alignment applied.